Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element B Initial Template

How to use this template

Element B: Documentation and Analysis of Prior Design Attempts

Name: ______________________________ **Date:** _________________ **Class:** _______________

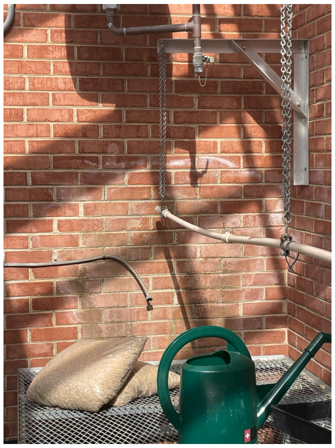
Prior Design Attempt #1 -

Source

Mold growth in greenhouse watering systems is a common issue that can adversely affect plant health. According to the U.S. Environmental Protection Agency (EPA), mold spores require moisture to grow, and controlling indoor moisture is essential to prevent mold proliferation.

"A Brief Guide to Mold, Moisture, and Your Home." *United States Environmental Protection Agency*, www.epa.gov/mold/brief-guide-mold-moisture-and-your-home. Accessed 25 Feb. 2025.

Image



Details : Covered in mold, PVC pipes have become bent and lost their previous straight shape, likely leading to uneven watering even if there was no mold.

Analysis : The prior watering design failed to keep outside contaminants from entering the watering system. One poor design was the fact that air would flow in and out of the system when the watering hose was not connected; This may have resulted in mold particles being able to enter the system when it wasn't running.

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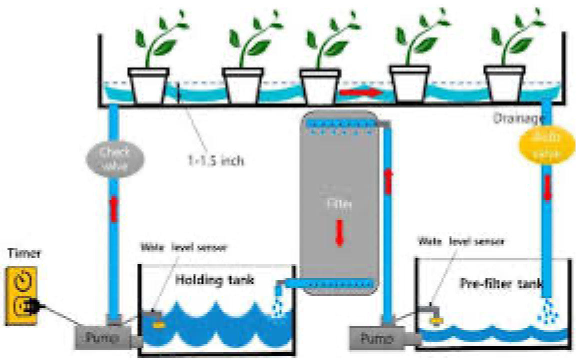
Prior Design Attempt #2 -

Source

In greenhouses, pathogens such as *Pythium* and *Phytophthora* can become problematic in hydroponic systems, especially when bottom sediment from water supplies is pumped into the irrigation system. These pathogens thrive in moist environments and can lead to significant plant diseases.

Chase, A. R. "Managing Botrytis or Gray Mold in the Greenhouse." *Penn State Extension*, extension.psu.edu/managing-botrytis-or-gray-mold-in-the-greenhouse. Accessed 25 Feb. 2025.

Image



Details: The system works on a timer, pumping the water up from the pre- filter holder, to the filter to the holding tank, up to the plants using a pump. It allows the plants to grow properly, on a timer and all evenly.

Analysis : In this design, we are able to see that the system can not get molded, based on the type of materials used. We can also see that the water keeps running and has a specific system, that way the water doesn't sit and end up molding or becomes moist in the pipes.

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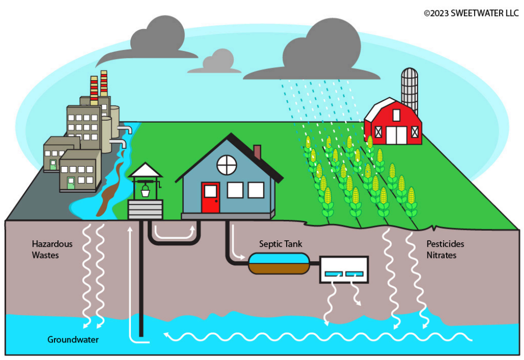
Prior Design Attempt #3 -

Source

To mitigate mold issues, it's crucial to manage moisture levels within the greenhouse. Ensuring proper drainage, avoiding overwatering, and maintaining appropriate humidity levels can help prevent mold growth. Regular monitoring and prompt attention to any signs of mold or moisture problems are essential steps in maintaining a healthy greenhouse environment.

"A Brief Guide to Mold, Moisture, and Your Home." *United States Environmental Protection Agency*, www.epa.gov/mold/brief-guide-mold-moisture-and-your-home. Accessed 25 Feb. 2025.

Image



Details : Here we can see that the rain is a system of giving water to the plants, but it's random of when it will come and can't be chosen. We also see that the system collects water from underwater and filters and goes up.

Analysis : We see that the way the system works, causes no mold, due to the moisture not building up underwater. We see that the way it filters is clean, using minerals and the pump plays a great system to help it go from underwater up to the surface.

Name: ____________________________ **Date:** ________________ **Class:** ______________

Rubric

Element B: Documentation and Analysis of Prior Design Attempts						
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
B1. Documentation of plausible prior attempts to solve the problem and/or related problems is drawn from __________.	a wide array of clearly-identified and consistently-credible sources	a variety of clearly-identified and consistently-credible sources	several—but not necessarily varied—clearly-identified and generally-credible sources	a limited number of sources, some of which may not be clearly identified and/or credible	only one or two sources that may not be clearly identified and/or credible	missing or minimal (a single source that is not clearly identified and/or credible)
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
B2. The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — __________.	is consistently clear, detailed, and supported by relevant data	is clear, generally detailed, and supported by relevant data	is generally clear and contains some detail and relevant supporting data	is overly general and contains little detail and/or relevant supporting data	is vague and is missing any relevant details and/or relevant supporting data	cannot be inferred from information intended as analysis of past and/or current attempts to solve the problem